

- 3) sketch a Combinational logic to indicate parasitic device capacitance in a 2 I/p NOR, NAND gate!

parasitic capacitance appears in two input NOR and NAND gate circuits using CMOS gate. In CMOS each transistor junction, gate oxide, and drain/source node contributes a parasitic capacitance. these add delay to switching and limit high-frequency performance.

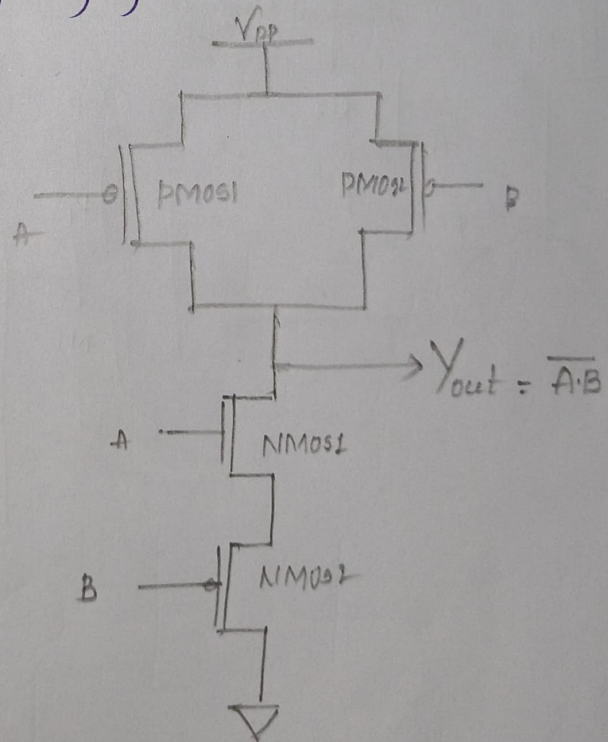
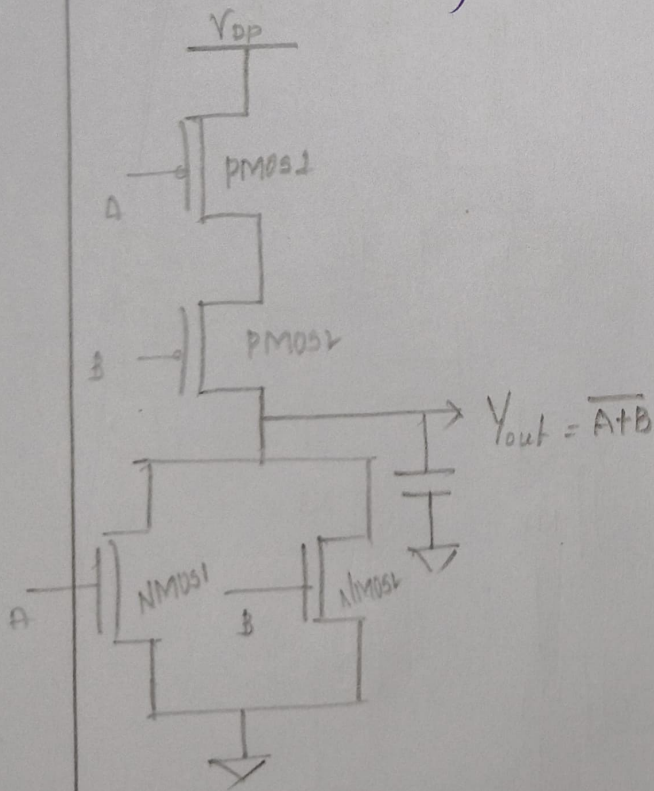
The key difference between two gates

- In the NOR gate both PMOS are in series and both NMOS are parallel so in pull up network one transistor connects directly to V_{DD} & both parallel NMOS will connect to ground.

parasitic capacitance:

output capacitance, internal node capacitance, Gate capacitance

- Internal capacitance shows pull-up-operation.

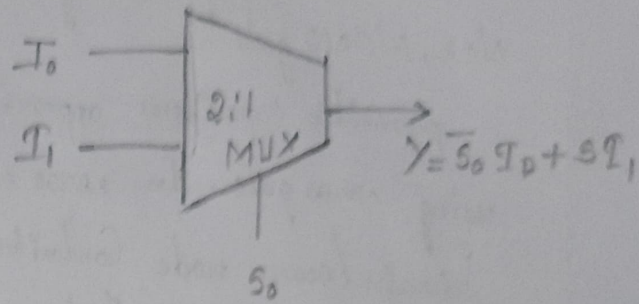
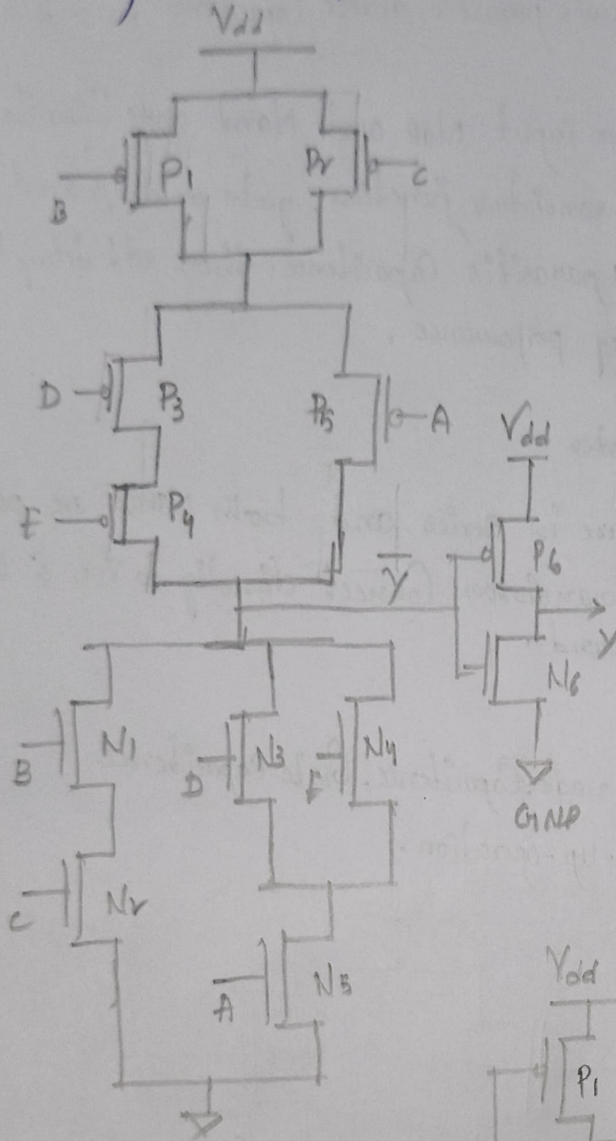


2) Realize the boolean functions using CMOS logic i) $[A(D+E)+BC]$ ii) 2 Input Mux!

Sol:

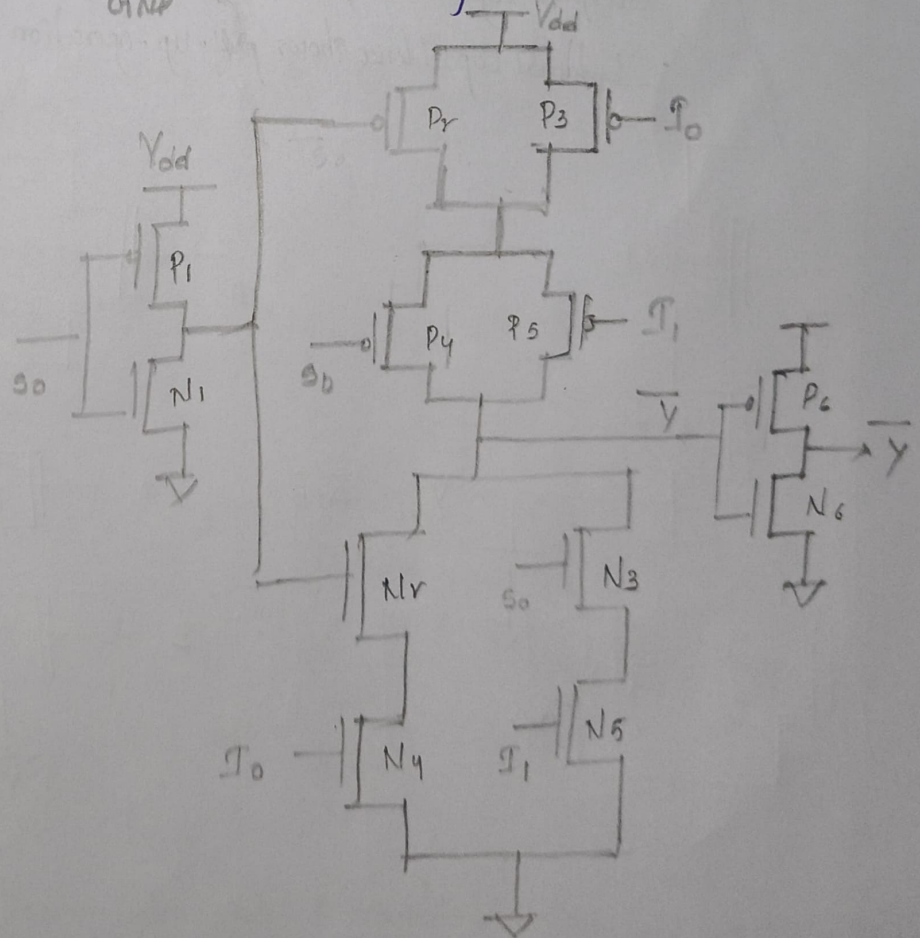
Given function $Y = [A(D+E)+BC]$

ii) 2 Input MUX



Input	output
0	I_0
1	I_1

$\therefore$ The expression $Y = \bar{S}_0 I_0 + S_0 I_1$

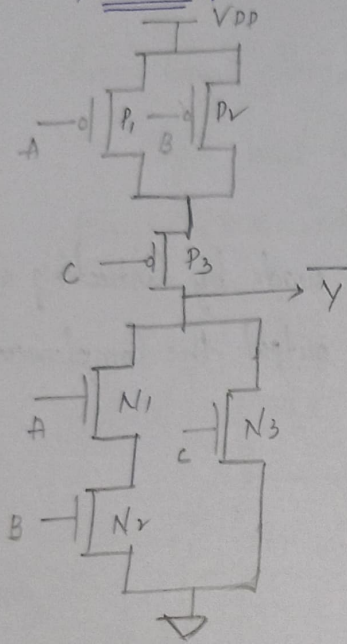


- 3) Analyze the operation of function $y = (AB + C)'$ using the logic design style provided
 i) CMOS 2) pseudo-NMOS 3) AOI logic 4) Domino logic.

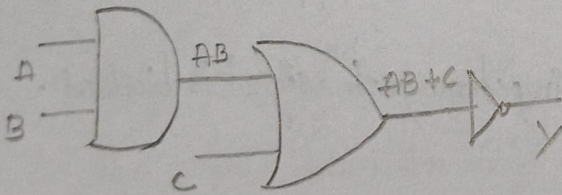
Ans:

Given logic = $y = (AB + C)'$

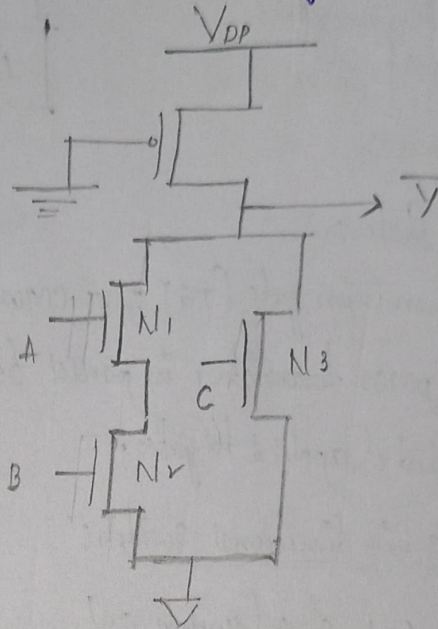
1) CMOS :



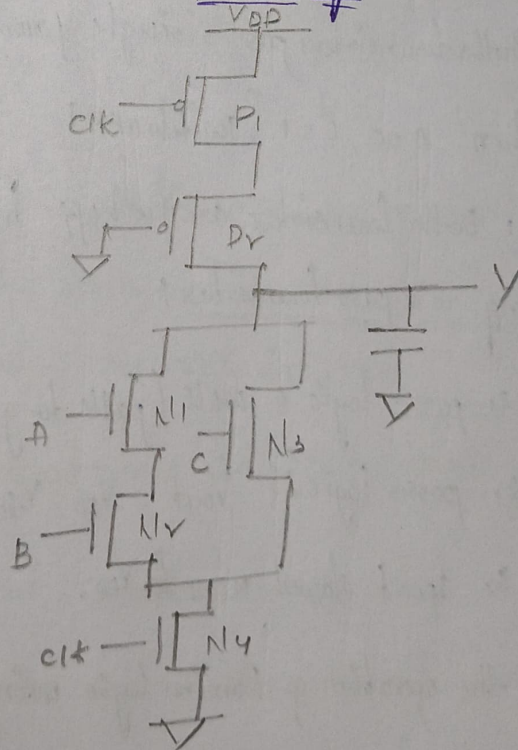
3) AOI :



2) Pseudo-NMOS logic



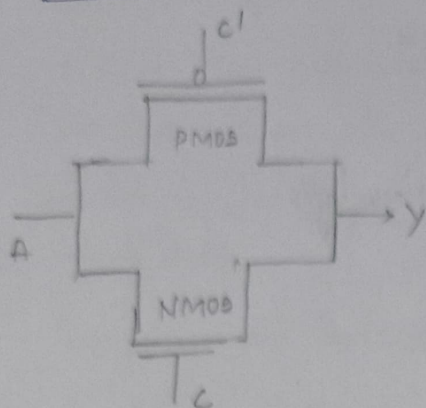
4) Domino logic



State the working of a transmission gate - is it a directional switch and pass transistor?

Ans:

Structure:



c	c'	state
1	0	ON
0	1	OFF

Transmission Gate:

A Transmission gate (TG) is a CMOS bilateral switch made by connecting an NMOS and a PMOS transistor in parallel between input and output the Complementary signal c and c' applied to gates.

Working as bi-directional switch:

When $c=1, c'=0$ (switch on)

• NMOS = high, PMOS = Low

Both transistor passes signal from A to Y or Y to A (truly bidirectional).

When $c=0, c'=1$ (switch off)

• Both transistor are cut off high impedance path signal is blocked.

Working as a pass transistor:

1, passes logic '0' well (pulls to GND).

2, passes logic '1' $V_{out} = V_{DD} - V_{th}$.

3, speed slower near to V_{DD} .

5) Analyze the operation of Domino logic using a Dynamic Nand gate?

operation:

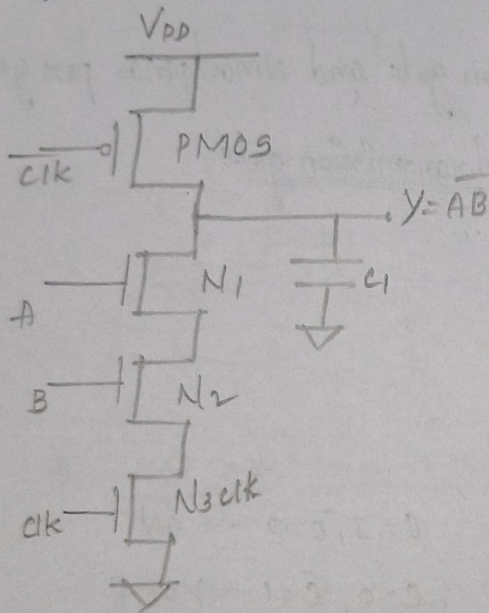
There are two phases operation in Dynamic

1, precharge

2, Evaluate.

Phase 1: Precharge

- $\text{clk} = 0 \rightarrow \overline{\text{clk}} = 1$ pmos $\rightarrow$ ON
- clk Nmos will off so the pull down network is Completely Disconnected from GND.
- Pmos charges the load capacitor $C_L V_{DD}$.
- $Y = \text{High}$ and $N_{\text{and output}} = 1$
- Input A & B are irrelevant during this phase.



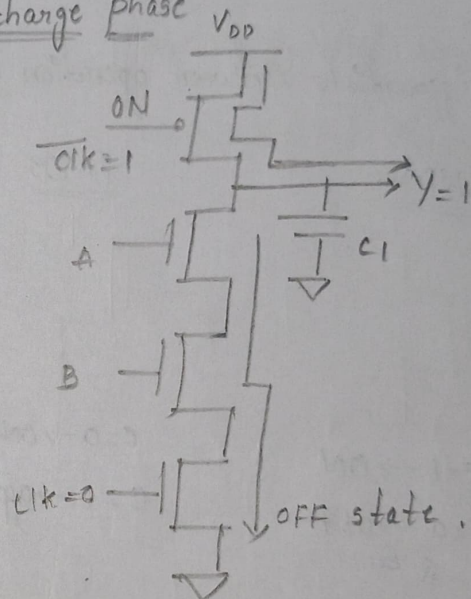
Phase 2: Evaluate

- $\text{clk} = 1 \rightarrow$ NMOS turn ON, PMOS OFF, so output depends entirely on A & B.

Case 1: A = 1, B = 1

- $N_1 \rightarrow$ ON, $N_2 \rightarrow$ ON, $\text{clk} = \text{ON}$ Complete Series path to GND.
- C_L - discharges $Q = 0$
- Output = 0

Precharge phase



- only 1 PMOS transistor is used & in static CMOS reduced area and capacitance.
- The output transition High $\rightarrow$ Low during evaluate, Low $\rightarrow$ High this is the domino property each gate can trigger the next in cascade.

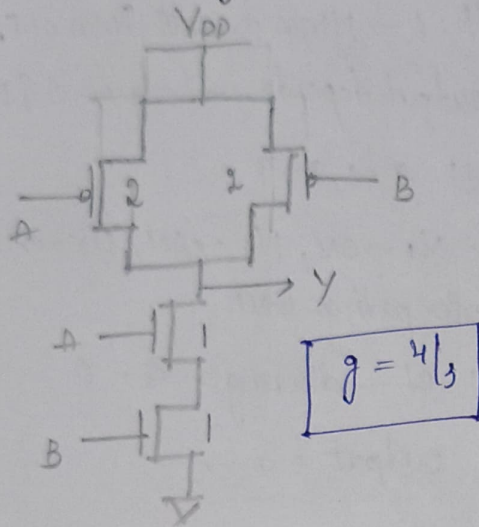
Sketch the logic to indicate logical effort in a 2 I/p NOR, NAND gate?

It is defined as the ratio of input capacitance of the gate to the input capacitance of the inverter that can deliver same output Current.

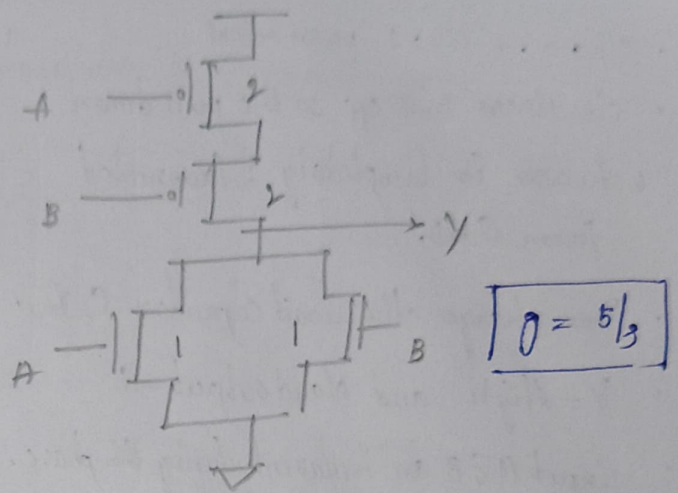
- In general n-input nand gate $g = \frac{n+2}{3}$

- In general n-input nor gate $g = \frac{2n+1}{3}$

1, Nand gate:



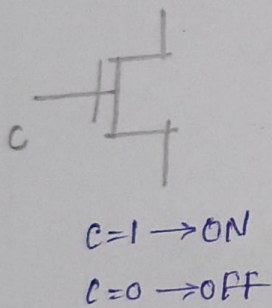
2, Nor gate:



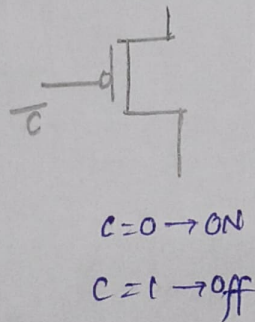
4, Differentiate Between operation of Transmission gate and NMOS/PMOS pass gate logic?

Ans

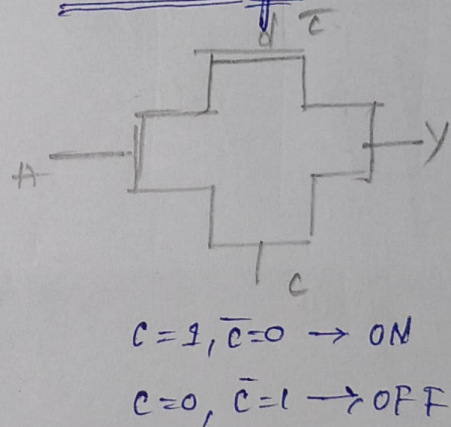
NMOS



PMOS



Transmission gate

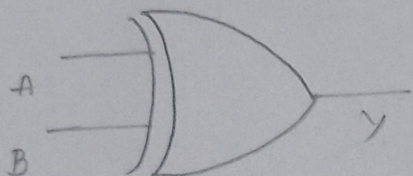


Features	NMOS	PMOS	Transmission gate
Transistor used	1 (NMOS)	1 (PMOS)	2 (NMOS + PMOS)
Control signal	C only	$\bar{C}$ only	Both C & $\bar{C}$
passes logic 1	weak	strong	strong
passes logic 0	strong	weak	strong
Speed	fast	fast	slow
Area	small	small	larger.
Bidirectional	Yes	Yes	Yes

2) Realize an EX-OR gate using pass transistor and Transmission gate?

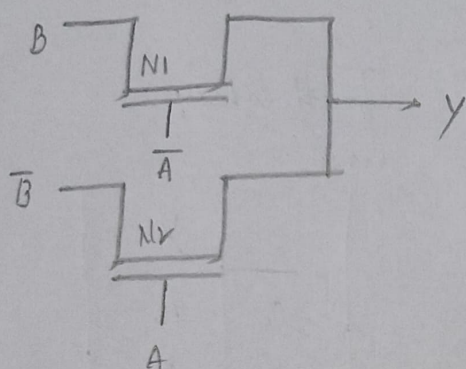
Sol: Given that XOR gate using pass transistor & Transmission gate

XOR: $\bar{A}B + A\bar{B}$



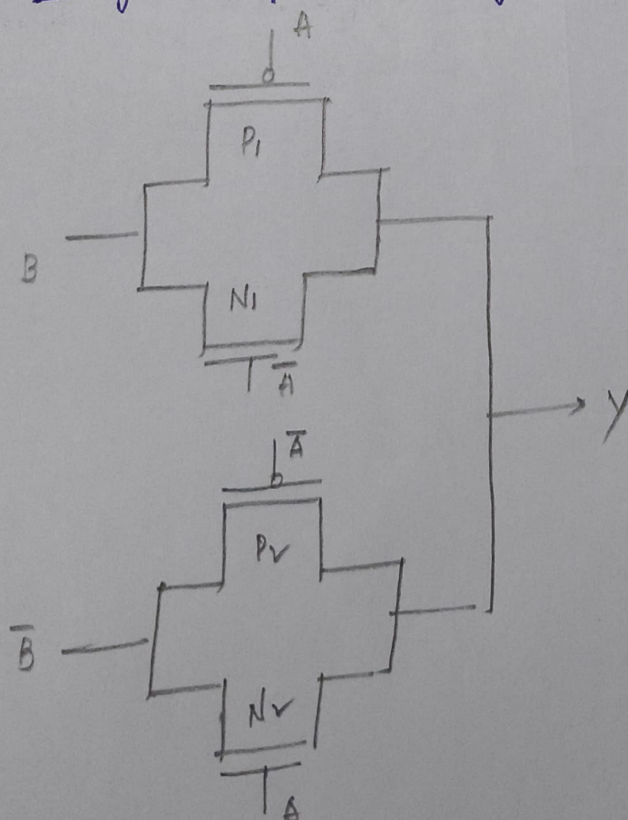
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

1, XOR gate using pass transistor



- one pin is called Controlled pin
- one pin is called Information pin

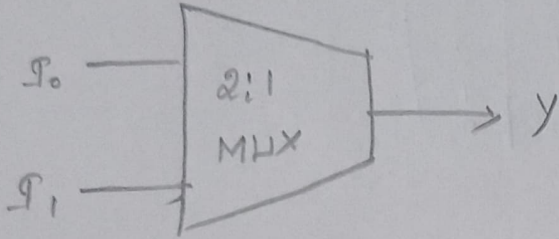
2, XOR gate using transmission gate



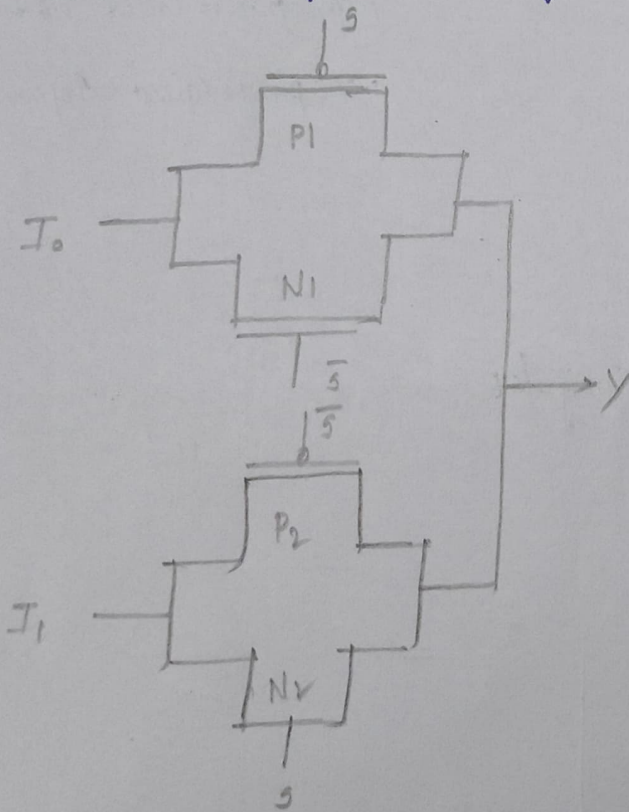
Q) Implement two-input multiplex circuit using two CMOS PMOS!

Ans: Given 2 input MUX $\rightarrow$ 2:1

Where $Y = \overline{S} I_0 + S I_1$



2:1 MUX using Transmission gate



Truth table:

S	Y
0	I_0
1	I_1